

Week 03

XML and JSON

Web Data Science

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Tuesday

Agenda

- 00:00 – 00:05 → Daily Note check-in
- 00:05 – 01:05 → Notebook lecture & exercises
- 01:05 – 01:15 → Daily note and feedback with Jonathan Featherstone

Daily Note questions

- Was there anything illegal about the Cambridge Analytica scraping?
 - In Europe? Maybe to probably. In the US? Probably not to maybe.
- Similar issues with data access with non-API data?
 - Data dumps used to be common (Wikipedia, StackOverflow, etc.), but also becoming more rare
- How to write a proposal in a slide deck?
 - Examples on Thursday!
- How to host a server for a friend?
 - Theoretically, `python -m http.server --bind 192.168.XXX.XXX` but you'll need to mess with router settings (disable DHCP, other security settings) and probably won't work on CU wifi
- Do I follow any sports teams?
 - I am a bandwagon Red Sox fan

Notebook

- Download “Class 03.ipynb” from Canvas and open in Jupyter Notebook
- Working with XML and JSON data formats
- Using libraries like requests and BeautifulSoup

Next class

- Each notebook will close with some exercises to take home
- Come to class on Thursday with some “Show-and-Tell” examples
 - Challenging bugs
 - Interesting data
 - Provocations for research designs
- Brainstorming about Module Assignment 01
- Daily note: <https://bit.ly/info4871f24note>
- Today’s number is... 2

Thursday

Agenda

- 00:00 – 00:05 → Daily Note check-in
- 00:05 – 00:20 → Wrap-up notebook
- 00:20 – 01:00 → Show & tell
- 01:00 – 01:10 → Module Assignment 01 brainstorming
- 01:10 – 01:15 → Wrap-up and daily note

Daily Note questions

- Most common data formats when web scraping?
 - HTML and JSON
- Why is Oasis the best band of all time?
 - Where were you while we were listening to Prince?
- Favorite hike in Boulder area?
 - Ceran Saint Vrain past Jamestown
- Harder examples?
 - Start low and go slow! Plenty of opportunities to ramp up difficulty over rest of semester

Module Assignment 01

- **Objective:** Write up a research proposal for analyzing the legislative activity of the Colorado Legislature.
- **Deliverable:**
Minimum five-page slide deck submitted as a PDF to Canvas by **Monday, September 23**
 - **Motivation:** What is the question or problem? Why is it important? Why could this data answer it?
 - **Data source:** What is the specific construct to measure? Screenshot a coleg.gov web page and circle it
 - **Methods:** What data will you need to collect to address your question? What will you do with the data?
 - **Expected results:** What kinds of analyses or visualizations will you do with the data to answer the RQ?
 - **Implications & Limitations:** What should Coloradans do if your RQ is right? What are limits to your analysis?
- **Evaluation:** More details in each will get higher scores. Screenshots, sketched visualizations, linking to scholarly or journalistic coverage, admitting limitations

FAQ

- Can we work together?
 - To some extent, but separate submissions needed but (near) identical submissions will be penalized
- I don't know anything about the Colorado General Assembly or state legislatures
 - <https://leg.colorado.gov/content/citizens-guide-effective-legislative-participation>
- What is an example of research questions about state legislatures?
 - Next two slides have 10 research questions from ChatGPT. [I dare you to do better.](#)
- This is a stupid idea and I want to do something else
 - I have an inbox full of journalists, politicians, and scientists who disagree. November 1 is drop date.

ChatGPT's 10 open research questions

How do term limits affect the legislative behavior and policy priorities of state legislators?

Do term-limited legislators behave differently in terms of voting patterns compared to those without term limits?

What role does partisanship play in shaping voting patterns in state legislatures, and how has this changed over time?

Are state legislatures becoming more polarized, and how does this affect bipartisan cooperation or policy outcomes?

How do state legislators balance the demands of their constituents with the interests of their political party?

When are legislators more likely to vote in line with their party versus the preferences of their constituents?

What impact do lobbyists and special interest groups have on the legislative behavior of state legislators?

How does the influence of money and what mechanisms allow lobbyists to shape policy outcomes?

How does the demographic composition of a state legislature affect its policy priorities and outcomes?

Does the presence of more diverse legislators influence the types of policies that are introduced and passed?

ChatGPT's 10 open research questions

What factors contribute to the legislative effectiveness of individual state legislators?

How do personal characteristics (e.g., seniority, political experience) and institutional factors (e.g., committee assignments, leadership positions) affect a legislator's ability to pass legislation?

How do state legislators' personal ideologies evolve over time, and what triggers shifts in their political behavior?

Are there particular events, pressures, or experiences that lead legislators to moderate or radicalize their views?

What are the consequences of increasing reliance on model legislation provided by national organizations like ALEC?

How does the use of model legislation affect the diversity of policy solutions proposed?

How do media coverage and public opinion shape the behavior and decision-making of state legislators?

Do state legislators respond to shifts in media framing and changes in public opinion?

How do gerrymandering or primary challenges influence the policy decisions of state legislators?

Do legislators take more extreme positions or avoid compromise when facing primary threats or redistricting?

Example MA 01 maybe earning a C+

Thanks ChatGPT 4o!

Motivation

- **Research Question:** How have legislative priorities in the Colorado General Assembly evolved over time?
- **Importance:** Understanding shifts in legislative focus can reveal policy trends, public concerns, and political dynamics in Colorado.
- **Relevance:** Legislative archives contain detailed records of bills, debates, and amendments, providing a rich source of historical data to track these changes.
- **Data Potential:** Analyzing this data can help identify patterns in lawmaking, highlight the influence of social, economic, or environmental issues, and inform future policy decisions.

Data source

- **Construct to Measure:** Legislative records, including bill summaries, amendments, and voting records.
- **Website:** The legislative archives available on the Colorado General Assembly website (leg.colorado.gov).
- **Focus:** Extract metadata (bill numbers, dates, keywords) and content (bill texts, voting outcomes) for analysis.

Methods

- **Data Collection:** Scrape bill summaries, full text, voting records, and metadata (e.g., sponsors, committees) from legislative archives.
- **Tools:** Use Python libraries like BeautifulSoup and Requests for scraping, store data in a structured format (e.g., CSV or database).
- **Analysis Plan:** Perform text analysis on bill content (e.g., topic modeling, keyword frequency), time series analysis of bill introduction, and voting patterns across different sessions.

Expected results

- **Visualizations:**
 - Time series showing trends in legislative focus (e.g., environmental policy, healthcare) across years.
 - Heat maps or bar charts displaying voting patterns by party, issue, or legislative session.
 - Word clouds or topic models to highlight key terms and concepts in bills over time.
- **Analyses:** Identifying correlations between external factors (e.g., elections, crises) and legislative output or priorities.

Implications & Limitations

- **Implications:** If successful, the analysis could provide valuable insights into how legislation reflects public concerns, influences policy discussions, and shapes Colorado's future.
- **Actionable Insights:** Data-driven understanding of past legislative trends could guide future policymaking and public engagement.
- **Potential Limitations:**
 - The analysis may be constrained by the quality and completeness of archived data.
 - Challenges in interpreting legislative intent and context from text data alone.
 - Biases in which topics are documented or prioritized over time.

Feedback for ChatGPT's MA01

- Missing details like:
 - Screenshots of what the data is
 - Sketched visualizations of what analyses could look like
 - Links to scholarly or journalistic coverage as motivation
- Hand-wavy research question → find an interesting motivation or gap!
- Hard to measure constructs → use specific examples!
- Trying to do too much → don't propose a Ph.D. dissertation!
- Tight relationship between motivation, source, methods, results, implications

Next week

- Review HTTP and HTML fundamentals, working on Module Assignment 01
- Daily note: <https://bit.ly/info4871f24note>
- Today's number is...